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To the Editors,
Econometrics and Statistics

Re: "An Exact Decomposition of Paired Score Differences by Relabelling Within Groups" — new submission

Dear Editors,

Please consider the enclosed manuscript for publication in *Econometrics and Statistics*, in the *Annals of Statistical Data Science* section. The paper is a methodological contribution to the comparison of probabilistic prediction models, and its applications are machine-learning benchmarks rather than economic or financial data, which is why I am directing it to that section rather than to Part A.

The problem, in one paragraph. Two probabilistic classifiers are compared on a test set and the newer one scores higher. If the labels depend partly on a grouping variable that both models observe, the score difference sums two distinct improvements: the new model may fit the label frequencies *within* groups more closely, or it may match its predictions to the individual cases they were made for more closely. These are different achievements — the first is skill a lookup table also supplies and that a shift in label frequencies removes; the second is not — and neither an aggregate score difference nor a significance test on it can tell them apart, since both components sit inside the same number.

The contribution. The paper decomposes the difference exactly, using no fitted reference model. The construction is elementary: because both models emit a full probability vector for every test case, each model's score can be recomputed against any label, so one can score a case's predictions against a *different* case's label from the same group. That relabelling preserves the group and its label frequencies while breaking the correspondence between a prediction and the case it was made for. The gain that survives is the *transported gain*; the gain that disappears is a *within-group covariance gain*. The resulting identity is exact in the observed sample, for any score, with no held-out fold and no tuning parameter, at a cost of $O(NK)$ in N cases and K labels.

The remainder is a U-statistic of order two, and it is not a new quantity: it equals the within-group covariance between the two models' predicted-probability difference and the label indicator, which is the object isolated by Yates's (1982) covariance decomposition of a proper score, here evaluated for a *pair* of forecasters instead of one. One correction relative to earlier drafts bears on how the contribution is framed, so I state it rather than leave it to be found: this remainder is *not* the resolution term of the classical reliability–resolution partition. Classical resolution depends on a forecast only through the partition its level sets generate, so it survives any monotone rescaling of forecast values; a covariance between those values does not. The two coincide only when both models are calibrated within groups. The paper states that relation precisely and adopts confidence matching as procedure rather than caveat. What the classical decomposition does not supply in either reading is inference for the pairwise contrast, and that is where the paper's technical content sits: the covariance identity holds for the whole Bregman-score family, so the quadratic and logarithmic scores are named cases of one theorem; the dataset-level estimator is asymptotically normal with a consistent cluster-robust sandwich variance; and at a trivial grouping variable the statistic reduces exactly to a clustered Diebold–Mariano/Giacomini–White test, which locates it precisely relative to comparative forecast evaluation. Three further exact results settle design choices that would otherwise be judgement calls — an $(n_c - 1)/n_c$ attenuation that is why no sample splitting is needed, an explicit between-group covariance term for what merging groups does, and a composition of the transported component that shows it is not a measure of prior fit — and a Cauchy–Schwarz sensitivity bound, in the tradition of Rosenbaum and Cinelli & Hazlett, limits how far an unrecorded grouping variable could move the result. On that last point the paper reports the bound against its own estimates rather than only in the abstract: for one appendix study an unrecorded covariate correlated at 0.061 would account for the whole reported gain, and I say so.

Validation, and one application in detail. Because the object is a measuring instrument, the paper validates it before using it: signal recovery, null calibration, interval coverage in the clustered small-group regime that stresses it, and discriminant-validity designs establishing what the two components do *not* separate. One of those is a genuine limitation rather than a strength — a monotone recalibration moves score between the components while adding no information — and the paper responds with a matching protocol that is part of the procedure rather than a caveat attached to it.

The main application audits two publicly released scene-graph checkpoints of a widely cited debiasing method, evaluated in both modes with the original authors' code, so the models are not ours and the claimed improvement is one the field already reports. Once confidence is matched on held-out images and the audit runs at the object-class-pair grouping, the proper-score difference is overwhelmingly transported under both the quadratic and the logarithmic score, although the two disagree on whether any covariance remainder survives (-0.00006 under the quadratic score, interval containing zero; $+0.04396$ under the log score, interval excluding it) — in either reading at least an order of magnitude smaller than the transported shift. That configuration qualifier appears wherever the claim does, because two of the five configurations reported for this same pair do not support it. Since the method's construction subtracts a context-only prediction, the finding characterises it rather than indicting it, and the paper is explicit that the ten-point mean-recall movement and the proper-score decomposition are in different units with no bridge between them.

Three further studies are reported in an appendix, including a case in which a model that loses on every aggregate measure is shown to discriminate individual cases significantly better than the model it appears to lose to.

Fit, and one thing I want to be straightforward about. The central contribution is an estimand together with its exact finite-sample and asymptotic inference theory. The benchmarks are where the instrument is exercised and validated, not the claim: no state-of-the-art result is asserted, and none is needed for the argument.

Corollary 3 places the work in the tradition of comparative predictive ability. I should say plainly, though, that the empirical evidence is drawn from image and text classification benchmarks rather than economic or financial forecasting, and the references reflect that: a referee expecting a forecasting application will not find one. My judgement is that the contribution is a statistical one for which the data-science section is the right home, and that adding a token macroeconomic example would strengthen the paper's appearance more than its substance — but if the editors disagree I would rather be told at this stage than have referees spend time on it. I have written the paper so that the contribution is legible before any notation appears — a plain statement of the identity in the introduction, a four-point worked example that exhibits the main properties proved later, and a table fixing the paper's vocabulary against its standard statistical counterparts. The same four points then exhibit a model-selection decision the decomposition gets right and the aggregate score gets wrong: a candidate whose entire gain is transported becomes *worse* than the model it replaces under a shift in label frequencies, while one whose gain is covariance is untouched.

All code, cached model outputs and scripts needed to reproduce every number, table and figure are publicly available at `https://github.com/vinhqdang/WAGER`, including a verification script that checks each of the 89 reported values against the committed results files.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. An earlier and substantially different version was submitted to another journal and declined; that submission is closed. I am the sole author and approve this submission, declare no competing financial or personal interests, and received no specific grant for this work. The manuscript includes a declaration of generative AI use in its preparation, in accordance with journal policy.

Thank you for your consideration.

Sincerely,
Quang-Vinh Dang